Aim: To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)

IDE: Google Colab

Theory:

To preprocess your text simply means to bring your text into a form that is predictable and analyzable for your task. A task here is a combination of approach and domain. Machine Learning needs data in the numeric form. We basically used encoding technique (Bag Of Word, Bi-gram, n-gram, TF-IDF, Word2Vec) to encode text into numeric vector. But before encoding we first need to clean the text data and this process to prepare (or clean) text data before encoding is called text preprocessing, this is the very first step to solve the NLP problems.

Tokenization:
Tokenization is about splitting strings of text into smaller pieces, or "tokens". Paragraphs can be tokenized into sentences and sentences can be tokenized into words.

Filtration:
Similarly, if we are doing simple word counts, or trying to visualize our text with a word cloud, stopwords are sonic of the most frequently occurring words but don't really tell us anything. We're often better off tossing the stopwords out of the text. By checking the Filter Stopwords option in the Text Pre-processing tool, you can automatically filter these words out.

Stemming:
Stemming is the process of reducing inflection in words (e.g. troubled, troubles) to their root form (e.g. trouble). The "root" in this case may not be a real root word, but just a canonical form of the original word.

Stemming uses a crude heuristic process that chops off the ends of words in the hope of correctly
actually be converted to troublinstead of trouble because the ends were just chopped off (ughh, how crude!). There are different algorithms for stemming. The most common algorithm, which is also known to be empirically effective for English, is Porters Algorithm. Here is an example of stemming in action with Porter Stemmer:

	original word	stemmed words
0	connect	connect
'I	connected	connect
2	connection	connect
3	connections	connect
4	connects	connect

Stopword Removal

Stop words are a set of commonly used words in a language. Examples of stop words in English are "a", ‘‘the", "is", "are" and etc. The intuition behind using stop words is that, by removing low information words from text, we can focus on the important words instead.

For example, in the context of a search system, if your search query is 'what is text preprocessing?", you want the search system to focus on surfacing documents that talk about text preprocessing over documents that talk about what is. This can be done by preventing all words from your stop word list from being analyzed. Stop words are commonly applied in search systems, text classification applications, topic modeling, topic extraction and others. Stop word removal, while effective in search and topic extraction systems, showed to be non-critical in classification systems. However, it does help reduce the number of features in consideration which helps keep your models decently sized

Program (Code):

```
text="""hello my name is Bhavin i am  currently in B.Tech ICT in marwadi university"""

#tokenization
text.split()
# i like playing cricket -4 words
# i like play ing cricket -5 tokens

import re #Regex: regular exspreesion
import string
string.punctuation

#nlp library
import spacy
#gensim,nltk

nlp=spacy.load('en_core_web_sm')
text
text[5]
doc=nlp(text)
doc
doc[11]

#NER => Name Entity Recognition
for token in doc.ents:
print(token.text,token.label_)

#post tags
for token in doc:
print(token.text,"->",token.pos_)
import nltk

#stopwords
```

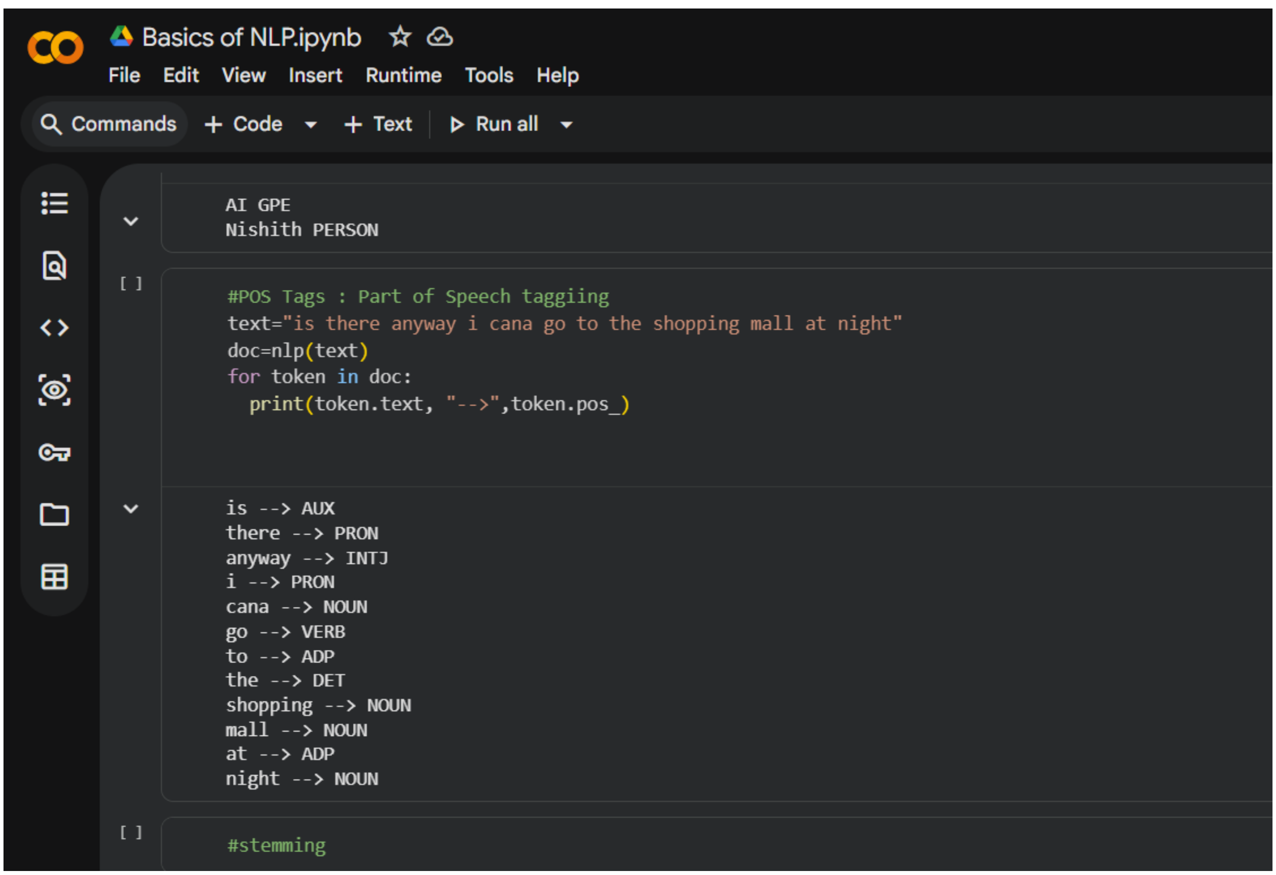
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)	
Experiment No: 6	**Date:**	**Enrolment No:92301733022**

```
from nltk.corpus import stopwords
nltk.download('stopwords')
set(stopwords.words('english'))

#Stemming vs lemmatization
from nltk.stem.porter import *
p_stem=PorterStemmer()
text=""" I study the subject name AI and then meeting Mr.Rohan tommarrow in
the meeting room"""
for word in text.split():
print(word,"->",p_stem.stem(word))

#lemmatization
for token in nlp(text):
print(token.text,"->",token.pos_,"->",token.lemma_)
```

Results:



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Commands + Code + Text Run all

```
#lemmatization
for token in nlp(text):
  print(token.text,"-> ",token.pos_,
        "->",token.lemma_)
```

```
I --> i
studied --> studi
AI --> ai
subject --> subject
and --> and
then --> then
meeting --> meet
Mr. --> mr.
Sachin --> sachin
in --> in
a --> a
meeting --> meet
I ->  PRON -> I
studied ->  VERB -> study
AI ->  PROPN -> AI
subject ->  NOUN -> subject
and ->  CCONJ -> and
then ->  ADV -> then
meeting ->  VERB -> meet
Mr. ->  PROPN -> Mr.
Sachin ->  PROPN -> Sachin
in ->  ADP -> in
a ->  DET -> a
meeting ->  NOUN -> meeting
```

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Observation:

From this experiment, we can clearly understand that text preprocessing is a crucial first step in any NLP task. Raw text data is usually unstructured, inconsistent, and contains unnecessary words or symbols. After applying preprocessing techniques, the text becomes more structured, clean, and suitable for further analysis or machine learning modeles

Post Lab Exercise:

1. Take your own document of 10 sentences and perform the same tasks. Attach code and screenshot of your output.

CODE:

```
# Install required libraries
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

# Sample document
text = """Natural language processing is a fascinating field.
Machines can understand human language with proper training.
Text preprocessing is the first step in NLP.
Tokenization splits text into words or sentences.
Stopwords are removed to improve efficiency.
Stemming reduces words to their base form.
Lemmatization gives meaningful root words.
Data cleaning improves model performance.
Python provides many libraries for NLP tasks.
Learning NLP is useful for many real-world applications."""

# Tokenization
sent_tokens = sent_tokenize(text)
word_tokens = word_tokenize(text)
print("Sentence Tokens:\n", sent_tokens)
print("\nWord Tokens:\n", word_tokens)

# Stopword Removal
```

```
stop_words = set(stopwords.words("english"))
filtered_words = [w for w in word_tokens if w.lower() not in stop_words]
print("\nFiltered Words:\n", filtered_words)

# Stemming
ps = PorterStemmer()
stemmed_words = [ps.stem(w) for w in filtered_words]
print("\nStemmed Words:\n", stemmed_words)

# Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(w) for w in filtered_words]
print("\nLemmatized Words:\n", lemmatized_words)
```

OUTPUT:

```
Sentence Tokens:
 ['Natural language processing is a fascinating field.', 'Machines can understand human language with proper training.', 'Text preprocessing is the first step in NLP.', 'Tokenization spl

Word Tokens:
 ['Natural', 'language', 'processing', 'is', 'a', 'fascinating', 'field', '.', 'Machines', 'can', 'understand', 'human', 'language', 'with', 'proper', 'training', '.', 'Text', 'preproces

Filtered Words:
 ['Natural', 'language', 'processing', 'fascinating', 'field', '.', 'Machines', 'understand', 'human', 'language', 'proper', 'training', '.', 'Text', 'preprocessing', 'first', 'step', 'N

Stemmed Words:
 ['natur', 'languag', 'process', 'fascin', 'field', '.', 'machin', 'understand', 'human', 'languag', 'proper', 'train', '.', 'text', 'preprocess', 'first', 'step', 'nlp', '.', 'token', '

Lemmatized Words:
 ['Natural', 'language', 'processing', 'fascinating', 'field', '.', 'Machines', 'understand', 'human', 'language', 'proper', 'training', '.', 'Text', 'preprocessing', 'first', 'step', 'N
```

```
nto words or sentences.', 'Stopwords are removed to improve efficiency.', 'Stemming reduces words to their base form.', 'Lemmatization gives meaningful root words.', 'Data cleaning improv

', 'the', 'first', 'step', 'in', 'NLP', '.', 'Tokenization', 'splits', 'text', 'into', 'words', 'or', 'sentences', '.', 'Stopwords', 'are', 'removed', 'to', 'improve', 'efficiency', '.',

'Tokenization', 'splits', 'text', 'words', 'sentences', '.', 'Stopwords', 'removed', 'improve', 'efficiency', '.', 'Stemming', 'reduces', 'words', 'base', 'form', '.', 'Lemmatization', 'g

ext', 'word', 'sentenc', '.', 'stopword', 'remov', 'improv', 'effici', '.', 'stem', 'reduc', 'word', 'base', 'form', '.', 'lemmat', 'give', 'meaning', 'root', 'word', '.', 'data', 'clean'

'Tokenization', 'split', 'text', 'word', 'sentence', '.', 'Stopwords', 'removed', 'improve', 'efficiency', '.', 'Stemming', 'reduces', 'word', 'base', 'form', '.', 'Lemmatization', 'give'
```

2. Write your observation for stemming and lemmatization you obtained for the sentences.

Stemming Observation

- Stemming removes suffixes using simple rules.
- Words are reduced to **non-meaningful roots** sometimes.
- Example:
    - *processing → process*
    - *fascinating → fascin* (not a proper word)
- It is **fast but less accurate**.

Lemmatization Observation

- Produces **meaningful base words** (dictionary form).
- Uses vocabulary and morphological analysis.
- Example:
    - *processing → processing* (or *process* depending on POS)
    - *better → good* (if POS applied)
- It is **slower but more accurate**.